SENIOR DATA SCIENTIST CANDIDATE ASSIGNMENT

NEURAL CONTENT RECOMMENDATION WITHOUT BEHAVIOUR DATA

EDA, cold-start evaluation and telecom production path

Kittituch Wongwatcharapaiboon (Top)

DATA CONSTRAINT

The catalogue supports content modelling. It contains no viewer behaviour.

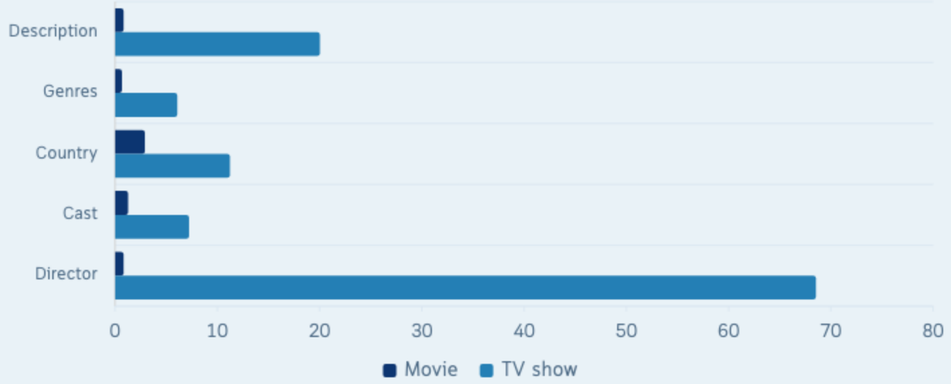
32,000
raw rows

31,991
titles after ID deduplication

1,991
deduplicated 2025 test candidates

0
users, impressions, watches or per-user ratings

METADATA MISSINGNESS (%)



DESIGN IMPLICATIONS

397 IDs collide across movie and TV files

68.5% of TV director values are missing

100% of movie duration values are missing

DESIGN DECISIONS

EDA changed the model, split and strength of the claim

EDA evidence	Decision	Why it matters
No customer interactions	Use content retrieval and label the evaluation as a metadata proxy	Avoids claiming collaborative filtering or customer lift
397 IDs collide across movie and TV; nine TV IDs repeat	Create type-prefixed IDs and retain the most complete duplicate	Prevents unrelated titles from sharing one item identity
The catalog is uniform by release year	Fit on 2010–2023, reserve 2024 and test unseen 2025 titles	Measures new-title cold start without fitting on the test period
Popularity and votes describe the current catalog	Exclude them from the learned retrieval embedding	Reduces temporal leakage and dependence on head titles
TV metadata has more missing values	Normalize fields and audit missingness by content type	Limits similarity driven by inconsistent source schemas

Evaluation: 500 episodes use three earlier titles to retrieve one held-out 2025 title from 1,991 candidates. The proxy tests metadata consistency, not customer lift.

ALGORITHMS AND WHY WE CHOSE THEM

Each method solves a different part of the catalog-only recommendation problem

Algorithm	Plain-language idea	Role in the pipeline	Status in final design
TF-IDF	Turns each title into a weighted word vector. Words that distinguish a title receive more weight.	Provides text features and a transparent benchmark	Input and baseline
Truncated SVD	Compresses 8,000 sparse word features into 256 dense latent concepts.	Reduces noise and makes neural training practical	Feature compression
Triplet neural encoder	Compares an anchor, a related title and an unrelated title. Training moves the related pair closer.	Creates item embeddings used by the inference retriever	**Selected retrieval model**
MMR reranking	Reorders the most similar candidates so the final list contains fewer near-duplicates.	Reranks retrieved candidates before delivery	**Selected reranker**



The neural encoder produces item embeddings during model inference and indexing. Online cosine search retrieves eligible candidates, and MMR reranks the final top 10.
Selected pipeline: Recall@10 47.8% and Coverage@10 66.8% on the metadata proxy.

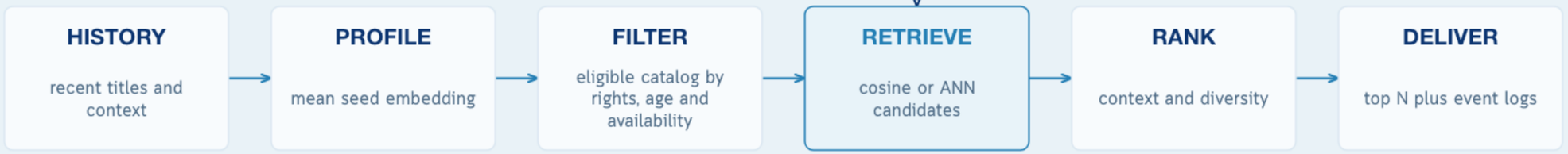
OFFLINE AND ONLINE ARCHITECTURE

Training publishes an item index; customer context enters during serving

OFFLINE TRAINING



ONLINE SERVING



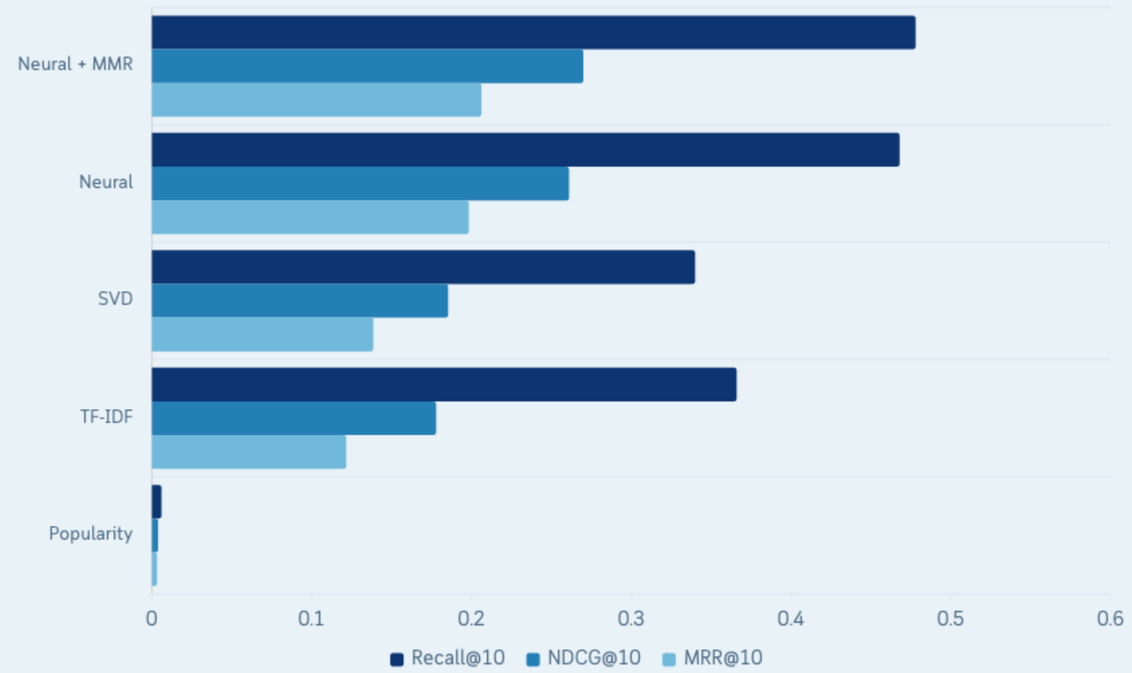
CURRENT ASSIGNMENT
Brute-force cosine across 1,991 candidates, followed by MMR

PRODUCTION EXTENSION
Eligible-catalog filtering, ANN retrieval, contextual ranking and behavior logs

RETRIEVAL BENCHMARK

Neural metric learning leads every ranking metric

COLD-START RANKING METRICS



+30.6%

Recall@10 versus TF-IDF

+51.8%

NDCG@10 versus TF-IDF

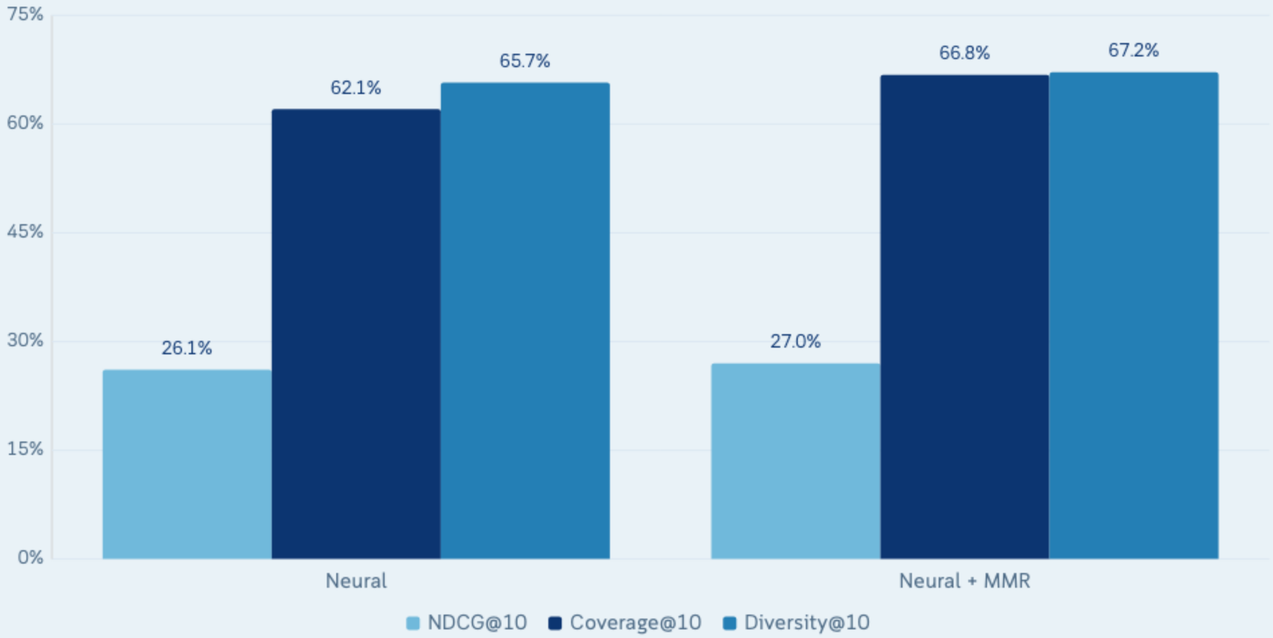
H1 SUPPORTED

Neural + MMR ranks the held-out target in the top 10 for 47.8% of episodes.

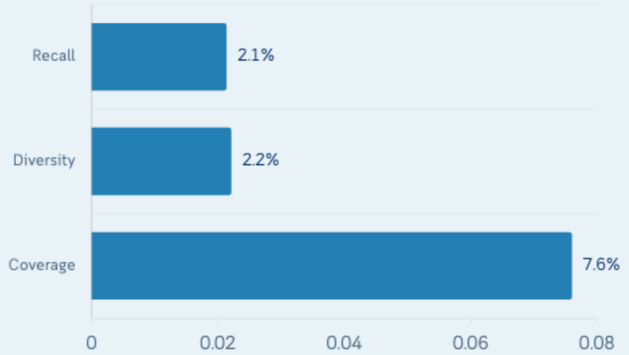
500 fixed-seed episodes. The result measures metadata consistency, not customer response.

DIVERSITY AND COVERAGE

MMR widens catalog reach and slightly improves ranking



RELATIVE MMR UPLIFT



+7.6%

catalog coverage versus the neural model without reranking

H2 remains partial: coverage improves, but the diversity threshold is not met.

MODEL SELECTION LOGIC

Each data constraint determines a design response and the claim we can support

	DATA CONSTRAINT	DESIGN RESPONSE	DECISION OR CONTROL
1	No customer interactions or impressions	Use content retrieval and label the evaluation as a metadata proxy	Avoid claims about collaborative filtering or customer lift
2	The target use case includes unseen 2025 titles	Fit on earlier metadata and hold out 2025 titles for the cold-start test	Measure unseen-item retrieval without fitting on the test period
3	Exact word overlap misses broader content relationships	Compress TF-IDF with SVD, then train a triplet neural encoder	Choose the retriever that leads Recall@10 and NDCG@10
4	Pure similarity can repeat near-duplicate titles	Retrieve by cosine similarity, then rerank the shortlist with MMR	Increase catalog coverage and list diversity

SELECTED DESIGN **Neural content retrieval with cosine ranking and MMR reranking**

Recall@10 47.8% NDCG@10 27.0% Coverage@10 66.8%

Best ranking quality in the proxy test, with broader catalog reach